

### Question 1:

With the help of truth table, verify whether the following statement (S) is a tautology or no.

- [3 points]  $S1: [(P \vee Q) \wedge (\neg P \vee R)] \rightarrow (Q \vee R)$
- [2 points]  $S2: (P \wedge (P \rightarrow Q)) \rightarrow Q$  What is this also called?
- [4 points] Convert the following statement to CNF and DNF both:  $(P \leftrightarrow (Q \vee R)) \wedge (\neg S \rightarrow T)$

V. [2 points] Suppose, these probabilities, that is, route  $i$  has traffic state 'High' depends upon the last two days traffic conditions on all the routes. Can we still model this as an MDP? If yes, what would be the states? If no, why?

**Question 3:**

- a. [2.5 points] I am storing a bit on my hard drive. I have changed my OS to return it in different way if a guest user tries to access it. This program simulates a fair coin toss and for Head, returns the original bit. If Tail, it again simulates a fair coin toss and if Head this time, returns 1 else 0. If probability of my original bit, say  $x$ , is one is  $p$ . That is,  $\Pr(x=1)=p$ . If you login to my computer using guest account and aware of the above program. If you read the bit to be 1, what is probability that the original bit is 1?
- b. [3 points] Recall our Meningitis example in the class. Let the events be as
- i. S: Proposition that patient has stiff neck
  - ii. H: Proposition that patient has severe headache
  - iii. M: Proposition that patient has meningitis

Assume S and H are independent given the patient has M.

Prove the following statement:

$$P(M | S \wedge H) = P(S | M) * P(H | M) * P(M) / P(S \wedge H)$$

- c. A certain disease affects 1% of the population. A test for the disease is 99% accurate, meaning it correctly identifies 99% of those with the disease (true positive rate) and correctly identifies 99% of those without the disease (true negative rate).
- i. [2 points] If a person tests positive, what is the probability that they actually have the disease?

- ii. [2.5 points] What is Accuracy, Precision and the Recall of the above test?
- d. [3 points] Suppose you have 6 data points in 2d with labels +1 or -1 as: (1,1), (2,2) and (2,0) belonging to +1 and (5,5), (5,6) and (6,5) belonging to the class -1. You recd new point (3,3) and you want to determine its label using kNN with  $k=1$  and  $k=3$ . What would be the labels respectively? Draw 2-dimensional diagram and mark the above points. Use Euclidian distances.

(.1)

10

1000

9.9

980.1

Consider the following data (as seen in the class)

5 of 5 columns

| Detail    | Compact | Column | a Temperature |   | a Humidity | a Wind | ✓ Play Tennis |
|-----------|---------|--------|---------------|---|------------|--------|---------------|
| a Outlook |         |        |               |   |            |        |               |
| Sunny     | 5       | 36%    | Mild          | 6 | 43%        | Weak   | 8             |
| Rain      | 5       | 38%    | Hot           | 4 | 29%        | Strong | 6             |
| Other (4) | 4       | 29%    | Other (4)     | 4 | 29%        |        |               |
| Sunny     |         |        | Hot           |   | High       | Weak   | No            |
| Sunny     |         |        | Hot           |   | High       | Strong | No            |
| Overcast  |         |        | Hot           |   | High       | Weak   | Yes           |
| Rain      |         |        | Mild          |   | High       | Weak   | Yes           |
| Rain      |         |        | Cool          |   | Normal     | Weak   | No            |
| Rain      |         |        | Cool          |   | Normal     | Strong | Yes           |
| Overcast  |         |        | Cool          |   | Normal     | Strong | Yes           |
| Sunny     |         |        | Mild          |   | High       | Weak   | No            |
| Sunny     |         |        | Cool          |   | Normal     | Weak   | Yes           |
| Rain      |         |        | Mild          |   | Normal     | Weak   | Yes           |
| Sunny     |         |        | Mild          |   | Normal     | Strong | Yes           |
| Overcast  |         |        | Mild          |   | High       | Strong | Yes           |
| Overcast  |         |        | Hot           |   | Normal     | Weak   | Yes           |
| Rain      |         |        | Mild          |   | High       | Strong | No            |

We need to make a decision tree to decide whether to play tennis or no. We would use CART algorithm step to split the tree. Here, splitting happens on the attribute which has least Ginni impurity rather than maximum information gain.

$$GI(S, A) = \sum_t \frac{s_t}{S} GI(S_t), \text{ where,}$$

- i.  $t \in T$  is the different values present in  $S$  for  $A$ ,
- ii.  $GI(S_t) = 1 - \sum_{i \in K} p_i^2$ ,  $K$  is the number of classes and  $p_i$  is probability of a random sample in class  $i$ . (we estimate it from the data, as we did it the class)

[12 points] Derive Decision tree for the above data.

Question 5:

[3+3 points] Please present set of steps for the Simulated Annealing search algorithm. You need to present two versions – first one is set of steps of the algorithm (for clarity of concept) and the second one is pseudo code (so finer or variable level details are captured).

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**Question 6:**

You trained an ML model that identified an  $n$ th order equation to fit a particular phenomenon. You plan to include regularization as part of the learning. There are two regularization functions (RFs) you designed: RF1 penalizes low order terms i.e., 1 to  $\lfloor n/2 \rfloor$  (i.e., floor of  $n/2$ ) while RF2 penalizes the rest of terms.

- a) [4 points] Please analyze and provide reasoning for what happens if we over penalize using RF1 and RF2 separately and provide their effect on the Bias and Variance in terms of whether they will increase, decrease or any other possibility for each of the cases ?

b) [2 points] What would be the effect on Bias and Variance if both RF1 and RF2 are used simultaneously? Please note you need to provide detailed reasoning for why you believe the effect you propose would happen?

... then

**Question 7:**

You are given the following D1 dataset  $(x, y)$  (where  $y = f(x) + \epsilon$ , noise  $\epsilon$  of 0 mean and variance  $\sigma^2$ ):  $(x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4), (x_5, y_5), (x_6, y_6), (x_7, y_7), (x_8, y_8), (x_9, y_9)$  corresponding to values  $(1, 3), (2, 6), (3, 9), (4, 12), (5, 15), (6, 18), (7, 21), (8, 24), (9, 27)$ . Using ML algorithm say MLA, you constructed ML model say MLM1 with the following model fit for the points  $\{(x_1, y_1'), (x_2, y_2'), (x_3, y_3'), (x_4, y_4'), (x_5, y_5'), (x_6, y_6'), (x_7, y_7'), (x_8, y_8'), (x_9, y_9')\}$  where the  $y'$  vector corresponds to  $\{y_1+2, y_2-1, y_3+2, y_4-1, y_5+2, y_6-1, y_7+2, y_8-1, y_9+2\}$ . Please answer the questions below along with all the steps and computations involved.

- a) [1+2 points] Please write down the formula for bias and explain the terms? Please compute bias for the information provided for model MLM1.

$$Bias = \hat{f}(x) - f(x) = E[f(x)]$$



### Question 8:

Below is a simplified Sudoku problem that involves a grid of 16 squares. The grid is divided into four blocks, each containing four squares. The rules of the problem are simple:

- a) The only digits allowed in the grid are 1, 2, 3 and 4
- b) Each of the four blocks has to contain all the numbers 1 to 4 within its squares.
- c) Each number between 1 to 4 can appear only once in a row, column or block.
- d) Each square must have a number filled.
- e) Notation: Rows are numbered A to D from top to bottom and columns are numbered 1 to 4 from left to right.

|   |   |   |   |
|---|---|---|---|
| 4 |   | 3 |   |
|   |   |   | 1 |
| 2 |   |   |   |
|   | 4 |   |   |

- a) [2+1+1 points] If you plan to solve the Sudoku problem using a (basic) hill climbing search what would your heuristic function be? What is the optimal value the algorithm should try to achieve with this heuristic? Please explain in detail?

- b) [4 points] Please show the first 4 steps of solution generated by (basic) hill climbing search using this heuristic? [Question aims to see your understanding of hill climbing -- you will therefore receive partial credit even if the heuristic above is not as well designed].

**Question 9:**

Consider a search space defined by table below, which gives the cost of arcs between pairs of nodes. Node A is the start state and node Z the goal state.

|      |   |   |   |   |   |   |   |   |   |
|------|---|---|---|---|---|---|---|---|---|
| From | A | A | A | B | C | D | E | F | G |
| To   | B | C | D | E | E | F | G | G | Z |
| Cost | 3 | 2 | 4 | 3 | 5 | 1 | 2 | 4 | 3 |

with the following heuristic function,  $h(n)$ :

|                  |   |   |   |   |   |   |   |   |
|------------------|---|---|---|---|---|---|---|---|
| node             | A | B | C | D | E | F | G | Z |
| $h(\text{node})$ | 8 | 4 | 6 | 6 | 3 | 6 | 2 | 0 |

|   |   |   |   |   |   |
|---|---|---|---|---|---|
|   | A | B | C | D | E |
| A | 0 | 3 | 2 | 4 |   |
| B | 8 | 4 | 6 | 6 |   |

- a) [2 points] Draw the complete graph of the state space [nodes and their connections].

b) [3 points] Using Uniform Cost Search, in what order would the nodes in the graph be expanded. Expanding a node means computing its successors and adding links to them. Please include clear explanation for the first 4 expansions, so we understand the logic behind your answer.

Expansion:  $C(2)$ ,  $B(3)$ ,  $D(4)$

- c) **[4 points]** Repeat this exercise using the selection function  $f(n) = g(n) + h(n)$ . Note that  $g(n)$  is the cost of the cheapest path identified from the start node to the node  $n$ . Please show trace of computation for first 4 steps in detail, so we understand the logic behind your answer.

Equation:  $B(7)$

d) [2 points] Is  $h(n)$  a good heuristic function to use -- why or why not ? Please explain.

- e) **[2 points]** If you were to propose a better heuristic  $h(n')$ , what would the heuristic be? Please explain why  $h(n')$  is better than the  $h(n)$  provided? If you cannot provide a better heuristic, you will need to argue why  $h(n)$  is the best possible heuristic.